








# Advanced Time Series Analysis and Forecasting

## Seminar 6: State space models and Bayesian filtering




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# Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **APF**: Auxiliary Particle Filter
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BN**: Beveridge–Nelson (decomposition)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CEE**: Central and Eastern Europe
- **DFM**: Dynamic Factor Model
- **DK**: Durbin–Koopman (simulation smoother, 2002)
- **ESS**: Effective Sample Size
- **FFBS**: Forward Filtering, Backward Sampling
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product

# Acronyms used in this material (2/2)

- **HICP**: Harmonised Index of Consumer Prices
- **IG**: inverse gamma (distribution)
- **IMA**: Integrated Moving Average (model)
- **KSC**: Kim–Shephard–Chib (1998), mixture sampler for stochastic volatility
- **LR**: Likelihood Ratio (test)
- **MA**: Moving Average
- **ML**: Machine Learning
- **MNZ**: Morley–Nelson–Zivot (2003)
- **PMMH**: Particle Marginal Metropolis–Hastings
- **QML**: Quasi-Maximum Likelihood
- **SCA**: Seasonally and Calendar Adjusted
- **SV**: Stochastic Volatility
- **TSA**: Time Series Analysis
- **TVP**: Time-Varying Parameter(s)
- **UC**: Unobserved Components (model)
- **UC-SV**: Unobserved Components model with Stochastic Volatility (Stock–Watson 2007)
- **US**: United States
- **VAT**: Value Added Tax

## Today's question and seminar structure

- The question of the seminar
  - **how do we estimate what we cannot observe (a level, a trend, a volatility, a gap), and how do we tell a real signal from the noise of the estimator?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

# Exercise map

| Task | Question | Type | Model |
|---|---|---|---|
| A1, A2 | the exact diffuse Kalman filter by hand; the concentrated likelihood | Solved, Proposed | A1 |
| A3, A4 | one backward-sampling step; the inverse-gamma Gibbs block | Solved, Proposed | A3 |
| A5, A6 | moments of the SV model; the QML measurement equation | Solved, Proposed | A5 |
| A7, A8 | one bootstrap particle-filter step; auxiliary weights with an outlier | Solved, Proposed | A7 |
| A9, A10 | the Beveridge–Nelson cycle; local level, IMA(1,1) and BN | Solved, Proposed | A9 |
| B1, B2 | Romanian trend growth; TVP regressions for Hungary and Poland | Solved, Proposed | B1 |
| B3, B4 | SV of EUR/RON: Gibbs, then the particle filter | Solved, Proposed | B3 |
| B5 | the Romanian output gap in real time | Proposed | B1, A10 |
| C1, C2 | is Romanian trend inflation anchored?; what is wrong in an AI answer? | Proposed | B3, A1–A10 |

- **[Solved]**: the full solution is in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it following the model, and the solution is discussed in class
- Part A: calculations and derivations on paper; Part B: real data, solved in the notebook, each ending with an interpretation question; Part C: open questions, including the critique of an AI answer

# Data used

| Series | Source | Frequency | Period |
|---|---|---|---|
| Romania: real GDP (SCA) | Eurostat namq_10_gdp | quarterly | 2000–2026 |
| HICP: Romania, Hungary, Poland, euro area | Eurostat prc_hicp_minr | monthly | 2001–2026 |
| EUR/RON reference rate | BNR | daily | 2016–2026 |

- In the notebook: `kalman_filter`, `kalman_smoother`, `sim_smoother_dk`, `ffbs`, `sv_gibbs`, `pf_sv`, `fit_llt`, `fit_tvp`, `fit_smooth_trend`, `ucsv_gibbs`; no account or key is needed

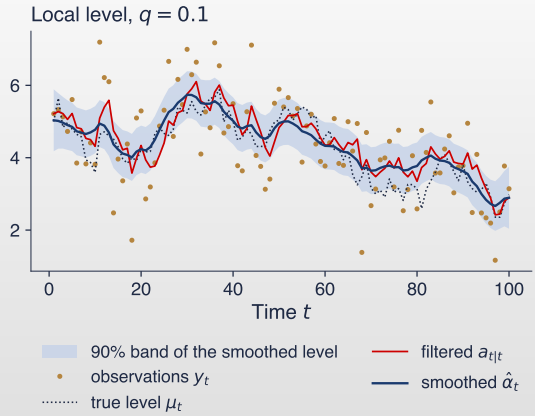
# Prerequisites for Today (1/19): the state space model

- Model (TSA, Chapter 10): observations are a noisy linear function of a hidden state that follows a linear dynamic

$$y_t = Z_t\alpha_t + \varepsilon_t,\ \varepsilon_t \sim N(0, H); \qquad \alpha_{t+1} = T\alpha_t + R\eta_t,\ \eta_t \sim N(0, Q)$$

- Notation
  - $y_t$: the observation; $\alpha_t$: the unobserved state; $Z_t$: measurement matrix; $T$: transition matrix; $R$: selects the states with shocks
  - $H$: variance of the measurement noise $\varepsilon_t$; $Q$: covariance of the state shocks $\eta_t$; $n$: the sample length
- Three estimates of the state, $Y_t = (y_1, \ldots, y_t)$
  - predicted $a_t = \mathbb{E}(\alpha_t \mid Y_{t-1})$, $P_t = \mathrm{Var}(\alpha_t \mid Y_{t-1})$; filtered $a_{t|t} = \mathbb{E}(\alpha_t \mid Y_t)$, $P_{t|t}$
  - smoothed $\hat{\alpha}_t = \mathbb{E}(\alpha_t \mid Y_n)$, $V_t = \mathrm{Var}(\alpha_t \mid Y_n)$: uses the whole sample
- Local level: $Z = T = R = 1$, $y_t = \mu_t + \varepsilon_t$, $\mu_{t+1} = \mu_t + \eta_t$; signal-to-noise ratio $q = \sigma_\eta^2/\sigma_\varepsilon^2$

# Prerequisites for Today (2/19): filtered and smoothed level




- Simulated local level, $n = 100$, $\sigma_\varepsilon^2 = 1$, $q = 0.1$
  - dots: the data; dotted line: the true level $\mu_t$
- Filtered $a_{t|t}$ (red): uses data up to $t$ only
  - what an analyst knows in real time; it reacts late
- Smoothed $\hat{\alpha}_t$ (blue): uses all $n$ observations
  - smoother and closer to $\mu_t$; band $\hat{\alpha}_t \pm 1.645\sqrt{V_t}$
  - at $t = n$ the two coincide

# Prerequisites for Today (3/19): the Kalman filter

- Prediction error and its variance

$$v_t = y_t - Z_t a_t, \qquad F_t = Z_t P_t Z_t' + H$$

  - $v_t$: the new information in $y_t$ (the innovation), independent of $Y_{t-1}$

- Update and prediction, with the Kalman gain $K_t = TP_t Z_t' F_t^{-1}$

$$a_{t|t} = a_t + P_t Z_t' F_t^{-1} v_t, \qquad a_{t+1} = Ta_t + K_t v_t, \qquad P_{t+1} = TP_t(T - K_t Z_t)' + RQR'$$

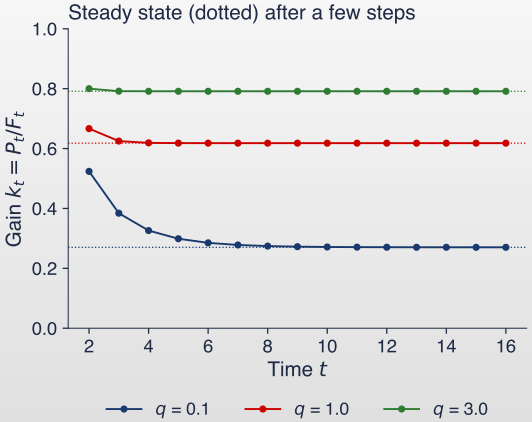
  - $K_t$: how much of the surprise $v_t$ moves the next state prediction; large when $P_t$ is large relative to $H$
  - local level: $a_{t+1} = a_t + k_t v_t$, $k_t = P_t/(P_t + \sigma_\varepsilon^2)$: an exponentially weighted average of the data

- Log-likelihood by the prediction error decomposition, a by-product of the filter

$$\ln L = -\frac{1}{2}\sum_{t=1}^{n}\left(\ln 2\pi + \ln F_t + v_t^2/F_t\right)$$

  - maximised numerically over the variances in $H$ and $Q$

# Prerequisites for Today (4/19): the gain and the smoother




- Local level, units of $\sigma_\varepsilon^2$: $P_{t+1} = P_t - P_t^2/(P_t + 1) + q$
  - $P_t$ and $k_t$ converge after a few steps (dotted lines): the steady state
  - a larger $q$: a larger gain, the level follows the data closely
- Smoother, backward from $t = n$

$$\hat{\alpha}_t = a_{t|t} + G_t(\hat{\alpha}_{t+1} - Ta_{t|t})$$

  - $G_t = P_{t|t}T'P_{t+1}^{-1}$: how much the next state tells about the current one
  - $V_t = P_{t|t} + G_t(V_{t+1} - P_{t+1})G_t'$

# Prerequisites for Today (5/19): the exact diffuse start

- Nonstationary states (random walks) have no stationary variance

$$P_1 = \kappa P_\infty + P_*, \qquad \kappa \to \infty$$

  - $P_\infty$: marks the diffuse elements; $P_*$: covariance of the others; $F_{\infty,t} = Z_t P_{\infty,t} Z_t'$ and $F_{*,t}$: the diffuse and finite parts of $F_t$

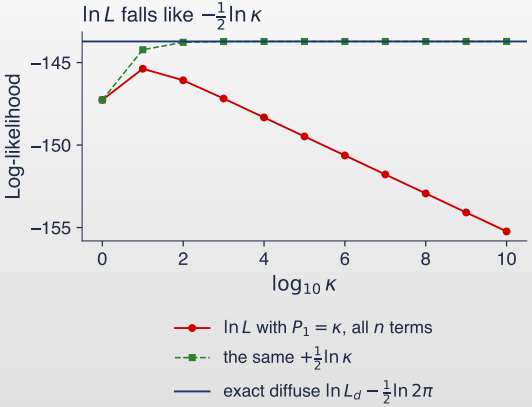
- While $F_{\infty,t} > 0$, two gains Koopman (1997)

$$K^{(0)} = TP_\infty Z'/F_\infty, \qquad K^{(1)} = T(P_* Z' - P_\infty Z' F_*/F_\infty)/F_\infty$$

  - $a_{t+1} = Ta_t + K^{(0)} v_t$, $P_{\infty,t+1} = TP_\infty L^{(0)\prime}$, $P_{*,t+1} = TP_\infty L^{(1)\prime} + TP_* L^{(0)\prime} + RQR'$
  - $L^{(0)} = T - K^{(0)} Z$, $L^{(1)} = -K^{(1)} Z$; after $d$ diffuse steps $P_\infty = 0$ and the ordinary filter takes over

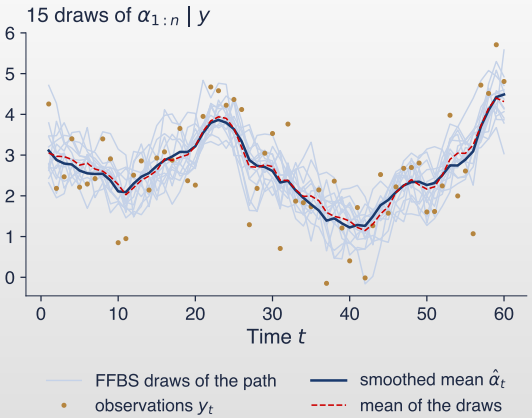
- Diffuse log-likelihood: a diffuse step contributes $-\frac{1}{2}(\ln 2\pi + \ln F_{\infty,t})$, the others the usual term
  - scale concentrated out: if $H$, $Q$ are proportional to $\sigma^2$, $\hat{\sigma}^2 = \frac{1}{n-d} \sum_{t>d} v_t^2/F_t^*$, $F_t^* = F_t/\sigma^2$

# Prerequisites for Today (6/19): a large $\kappa$ against the exact diffuse start




- Simulated local level, $n = 80$, $q = 0.5$
  - red: $a_1 = 0$, $P_1 = \kappa$ and all $n$ terms of $\ln L$
- The first term contains $-\frac{1}{2}\ln(\kappa + \sigma_\varepsilon^2)$
  - $\ln L \to -\infty$ as $\kappa$ grows; round-off errors grow too
  - adding back $\frac{1}{2}\ln\kappa$ (green) recovers the exact diffuse value (blue)
- Use the exact diffuse filter and drop the $d$ diffuse terms

# Prerequisites for Today (7/19): sampling the states




- FFBS Carter and Kohn (1994): forward filter, backward sampling
  - draw $\alpha_n \sim N(a_{n|n}, P_{n|n})$
  - then, for $t = n-1, \ldots, 1$: $\alpha_t \mid \alpha_{t+\mathbf{1}} \sim N(m_t, S_t)$

$$m_t = a_{t|t} + G_t(\alpha_{t+\mathbf{1}} - Ta_{t|t}), \quad S_t = P_{t|t} - G_t T P_{t|t}$$

  - $G_t = P_{t|t} T' (T P_{t|t} T' + RQR')^{-\mathbf{1}}$
- Durbin–Koopman Durbin and Koopman (2002): simulate $(\alpha^+, y^+)$, smooth once

$$\tilde{\alpha} = \alpha^+ + \hat{\alpha}(y) - \hat{\alpha}(y^+)$$

  - $\hat{\alpha}(\cdot)$: the smoothed mean; $\tilde{\alpha}$: an exact draw of the whole path
- Chart: the draws scatter around $\hat{\alpha}_t$; their mean is $\hat{\alpha}_t$

# Prerequisites for Today (8/19): Gibbs blocks for the variances

- Inverse gamma $IG(a, b)$: $\sigma^2 \sim IG(a, b)$ if $1/\sigma^2 \sim$ Gamma with shape $a$ and rate $b$

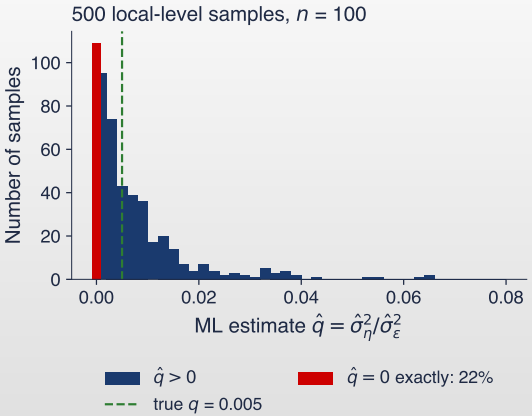
$$p(\sigma^2) \propto (\sigma^2)^{-a-1} e^{-b/\sigma^2}, \qquad \mathbb{E}\sigma^2 = \frac{b}{a-1} \ (a > 1)$$

  - $a$: shape; $b$: scale of $\sigma^2$ (the rate of $1/\sigma^2$); the density is zero at $\sigma^2 = 0$
- Gibbs block: prior $IG(a_0, b_0)$, $m$ shocks $e_t$ implied by the current state draw

$$\sigma^2 \mid e \sim IG\Big(a_0 + \frac{m}{2},\ b_0 + \frac{1}{2}\sum_t e_t^2\Big)$$

  - local level: $e_t = y_t - \mu_t$ for $\sigma_\varepsilon^2$ $(m = n)$, $e_t = \Delta\mu_t$ for $\sigma_\eta^2$ $(m = n-1)$
  - the sampler alternates: states given variances (FFBS), variances given states
- Inefficiency factor $1 + 2\sum_{k\geq 1} \rho_k$
  - $\rho_k$: autocorrelation of the chain at lag $k$; the number of correlated draws worth one independent draw

# Prerequisites for Today (9/19): variances at the boundary




- Pile-up Shephard and Harvey (1990): the ML estimate of a variance can be exactly 0
  - 500 local-level samples, $n = 100$, true $q = 0.005$
  - 22% of the estimates are exactly $\hat{q} = 0$
- LR test of $\sigma^2_\eta = 0$: the null is on the boundary
  - $LR = 2(\ln L_1 - \ln L_0) \geq 0$; standard $\chi^2_1$ is wrong
  - rule of thumb: $p = \frac{1}{2} P(\chi^2_1 > LR)$ (mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$)
  - random-walk alternatives: nonstandard; use a parametric bootstrap

# Prerequisites for Today (10/19): local trends and time-varying coefficients

- Local linear trend

$$y_t = \mu_t + \varepsilon_t, \qquad \mu_{t+1} = \mu_t + \nu_t + \xi_t, \qquad \nu_{t+1} = \nu_t + \zeta_t$$

  - $\nu_t$: the slope; $\xi_t$, $\zeta_t$: level and slope shocks with variances $\sigma^2_\xi$, $\sigma^2_\zeta$; two diffuse states
  - $y_t = 100 \ln$ of a quarterly series: $4\nu_t$ is the annualised trend growth, in %

- TVP (time-varying parameter) regression

$$y_t = c_t + b_t x_t + D_t'\gamma + \varepsilon_t, \qquad c_{t+1} = c_t + \eta^c_t, \quad b_{t+1} = b_t + \eta^b_t$$

  - state $\alpha_t = (c_t, b_t, \gamma')'$, $Z_t = (1, x_t, D_t')$; $D_t$: seasonal dummies, $\gamma$: constant states
  - $\sigma^2_b = \mathrm{Var}(\eta^b_t) = 0$: a constant slope; the smoothed $b_t$ shows when the link changed

# Prerequisites for Today (11/19): stochastic volatility

- SV model: the log-variance is a latent AR(1) with its own shock

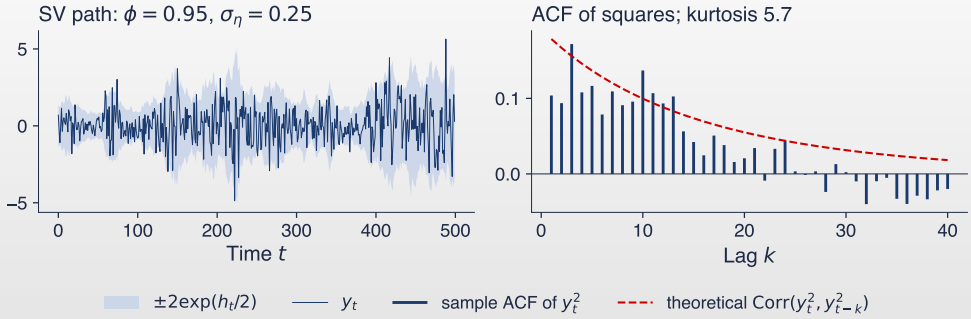
$$y_t = e^{h_t/2}\epsilon_t, \qquad h_t = \mu + \phi(h_{t-1} - \mu) + \sigma_\eta \eta_t, \qquad \epsilon_t, \eta_t \sim N(0,1)$$

  - $h_t$: log-variance; $\mu$: its mean; $\phi$: persistence; $\sigma_\eta$: volatility of volatility
  - half-life of a shock to $h_t$: $\ln 0.5 / \ln \phi$ periods
- Moments, with $\sigma_h^2 = \sigma_\eta^2/(1-\phi^2)$ the variance of $h_t$

$$\text{kurtosis} = 3e^{\sigma_h^2}, \qquad \text{Corr}(y_t^2, y_{t-k}^2) = \frac{e^{\sigma_h^2}\phi^k - 1}{3e^{\sigma_h^2} - 1}$$

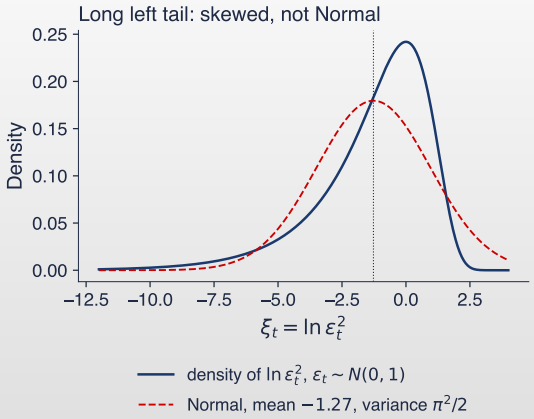
  - kurtosis above 3 and positive, slowly decaying autocorrelation of $y_t^2$: volatility clustering
- GARCH(1,1) Bollerslev (1986), for comparison: $\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2$
  - the variance is known one step ahead; in SV it has its own shock, so the likelihood is an integral over $h_{1:n}$

# Prerequisites for Today (12/19): a simulated SV path




- $\phi = 0.95$, $\sigma_\eta = 0.25$: $\sigma_h^2 = 0.0625/0.0975 = 0.64$, kurtosis $3e^{0.64} = 5.7$, half-life $\ln 0.5 / \ln 0.95 = 13.5$ periods
- Left: calm and turbulent spells inside the band $\pm 2e^{h_t/2}$; right: the ACF of $y_t^2$ starts low (0.18) and decays like $\phi^k$

# Prerequisites for Today (13/19): the linearised SV model




- Square and take logs

$$\ln y_t^2 = h_t + \xi_t, \qquad \xi_t = \ln \epsilon_t^2$$

  - $\mathbb{E}\xi_t = -1.2704$, $\mathrm{Var}\xi_t = \pi^2/2 = 4.93$
  - linear in $h_t$: a state space model
- QML Harvey, Ruiz and Shephard (1994): the Kalman filter as if $\xi_t$ were Normal
  - best linear predictor, but $\xi_t$ is skewed (chart)
- KSC Kim, Shephard and Chib (1998): a mixture of seven Normals

$$p(\xi_t) \approx \sum_{j=1}^{7} q_j \, N(\xi_t;\, m_j, v_j)$$

  - $q_j$, $m_j$, $v_j$: fixed weights, means, variances
  - given the component $s_t$, $h$ is Gaussian: FFBS applies

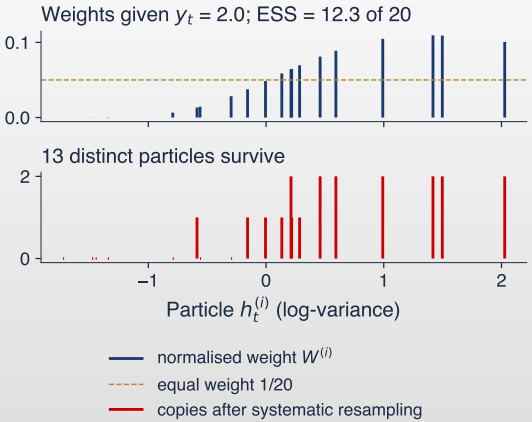
# Prerequisites for Today (14/19): particle filters

- Bootstrap particle filter Gordon, Salmond and Smith (1993): $N$ particles $x_t^{(i)}$ approximate the distribution of the state
  - propagate $x_t^{(i)} \sim f(\cdot \mid x_{t-1}^{(i)})$ ($f$: transition density); weight $w^{(i)} = g(y_t \mid x_t^{(i)})$ ($g$: measurement density)
  - normalised weights $W^{(i)} = w^{(i)} / \sum_j w^{(j)}$; then resample
- Likelihood estimate and the effective sample size

$$\hat{p}(y_t \mid Y_{t-1}) = \frac{1}{N} \sum_{i=1}^{N} w^{(i)}, \qquad \hat{L} = \prod_{t=1}^{n} \hat{p}(y_t \mid Y_{t-1}), \qquad \mathrm{ESS}_t = \frac{1}{\sum_i (W^{(i)})^2}$$

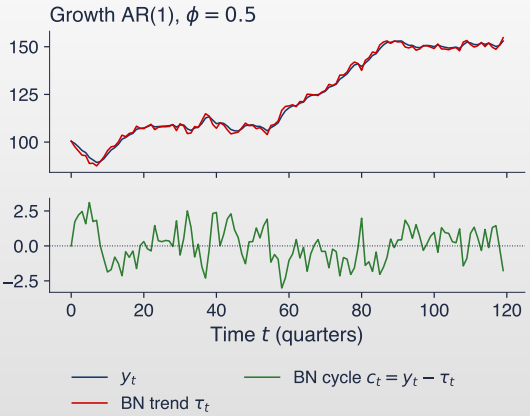
  - $\hat{L}$ is unbiased: $\mathbb{E}\hat{L} = L$; by Jensen, $\mathbb{E} \ln \hat{L} < \ln L$: the log is biased downwards
  - $\mathrm{ESS}_t$ between 1 and $N$; a collapse to a few particles: the model or the proposal fails at $t$
- Systematic resampling: one uniform $U$, $u_i = (U + i - 1)/N$
  - particle $j$ is copied once for every $u_i \in (C_{j-1}, C_j]$; $C_j = \sum_{l \le j} W^{(l)}$: the cumulative weights

# Prerequisites for Today (15/19): one step of a particle filter




- SV model, 20 particles $h_t^{(i)}$, observation $y_t = 2.0$
  - $w^{(i)} = N(y_t; 0, e^{h_t^{(i)}})$: a large $|y_t|$ favours high-volatility particles
  - ESS = 12.3 of 20
- Systematic resampling (bottom)
  - low-volatility particles die, high ones are copied twice
  - 13 distinct particles survive
- Auxiliary filter (APF) Pitt and Shephard (1999): look at $y_t$ before propagating
  - first-stage weights $\propto g(y_t \mid \mu_t^{(i)})$, $\mu_t^{(i)} = \mathbb{E}(x_t \mid x_{t-1}^{(i)})$
  - then propagate and correct the weights

# Prerequisites for Today (16/19): the Beveridge–Nelson decomposition




- BN trend Beveridge and Nelson (1981): the level once predictable growth dies out

$$\tau_t = y_t + \sum_{j \geq \mathbf{1}} \mathbb{E}_t(\Delta y_{t+j} - \mu)$$

  - $\mu$: mean growth; $\mathbb{E}_t$: expectation given data up to $t$
- AR(1) growth: $\Delta y_t - \mu = \phi(\Delta y_{t-\mathbf{1}} - \mu) + e_t$
  - cycle $c_t = y_t - \tau_t = -\frac{\phi}{\mathbf{1}-\phi}(\Delta y_t - \mu)$
  - chart, $\phi = 0.5$: $c_t = -(\Delta y_t - \mu)$, noisy, small
- Local level: $\Delta y_t = u_t + \theta u_{t-\mathbf{1}}$, an IMA(1,1)
  - $u_t$: white noise; $\theta = -(1 + q/2) + \sqrt{q + q^{\mathbf{2}}/4}$

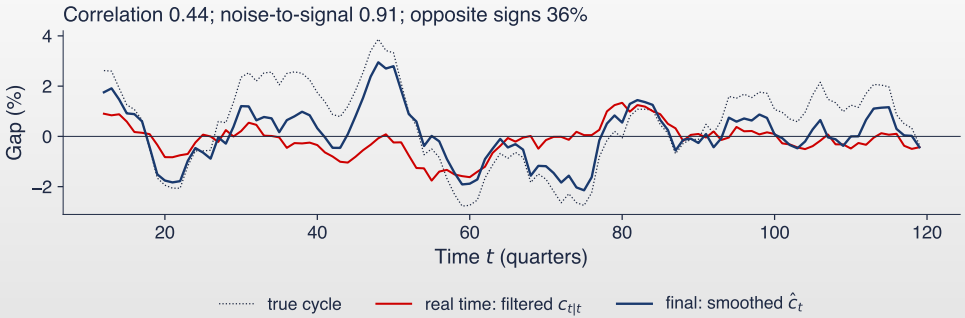
# Prerequisites for Today (17/19): output gaps in real time

- Smooth-trend UC (unobserved components) model with an AR(2) cycle

$$y_t = \tau_t + c_t, \qquad \Delta^2 \tau_{t+1} = \zeta_t, \qquad c_t = \phi_1 c_{t-1} + \phi_2 c_{t-2} + \kappa_t$$

  - $\tau_t$: potential output; $c_t$: the output gap; $\zeta_t$, $\kappa_t$: independent shocks; $\Delta^2$: second difference
  - real-time gap: the filtered $c_{t|t}$ from data up to $t$; final gap: the smoothed $\hat{c}_t$
- Orphanides–van Norden statistics Orphanides and van Norden (2002), revision $r_t = c_{t|t} - \hat{c}_t$
  - $\mathrm{Corr}(c_{t|t}, \hat{c}_t)$; noise-to-signal ratio $\mathrm{sd}(r_t)/\mathrm{sd}(\hat{c}_t)$, near 1 or above: unreliable
  - share of opposite signs: $\frac{1}{n}\sum_t 1\{\mathrm{sign}(c_{t|t}) \neq \mathrm{sign}(\hat{c}_t)\}$

# Prerequisites for Today (18/19): real-time against final gaps




- Simulated smooth trend plus AR(2) cycle ($\phi_1 = 1.4$, $\phi_2 = -0.5$), true parameters known: only the end-point problem is at work
- The filtered gap is flatter and often of the wrong sign: correlation 0.44, noise-to-signal 0.91, opposite signs in 36% of the quarters

# Prerequisites for Today (19/19): trend inflation: UC-SV

- UC-SV Stock and Watson (2007): inflation = trend + transitory noise, both with stochastic volatility

$$\pi_t = \tau_t + \sigma_{\eta,t}\eta_t, \qquad \tau_t = \tau_{t-1} + \sigma_{\varepsilon,t}\varepsilon_t$$

  - $\eta_t, \varepsilon_t \sim N(0,1)$, independent; $\tau_t$: trend inflation
  - $\ln \sigma^2_{\eta,t} = \ln \sigma^2_{\eta,t-1} + \nu_{\eta,t}$, $\ln \sigma^2_{\varepsilon,t} = \ln \sigma^2_{\varepsilon,t-1} + \nu_{\varepsilon,t}$, with $\mathrm{Var}\nu = \gamma$
  - $\gamma$: smoothness of the volatilities, fixed in advance (0.2 in Stock–Watson)
- Estimation by Gibbs: FFBS for $\tau$; KSC steps for the two log-volatilities
  - posterior probability of an event from the $M$ draws $\tau_t^{(m)}$: $P(1.5 \le \tau_t \le 3.5 \mid Y_n) \approx \frac{1}{M}\sum_m 1\{1.5 \le \tau_t^{(m)} \le 3.5\}$
  - a 68% band: the 16th and 84th percentiles of the draws

# A1: the exact diffuse filter by hand [Solved]

## Task

- Local level, $\sigma_\varepsilon^2 = 1$, $\sigma_\eta^2 = 0.5$, $\mu_{\mathbf{1}}$ diffuse ($a_{\mathbf{1}} = 0$, $P_{\infty,\mathbf{1}} = 1$, $P_{*,\mathbf{1}} = 0$); data $y = (2.0; 3.0; 2.5)$.
- 1. Run the diffuse step at $t = 1$: give $F_\infty$, $K^{(\mathbf{0})}$, $K^{(\mathbf{1})}$, $a_{\mathbf{2}}$, $P_{\infty,\mathbf{2}}$, $P_{*,\mathbf{2}}$.
- 2. Run the ordinary filter for $t = 2, 3$: $v_t$, $F_t$, $K_t$, $a_{t+\mathbf{1}}$, $P_{t+\mathbf{1}}$.
- 3. Compute the diffuse log-likelihood and say why $y_{\mathbf{1}}$ carries no information about the variances.
- Report: one diffuse step, two ordinary steps, one number, one sentence.

## Solution

- 1. $F_\infty = 1$, $F_* = 0 + 1 = 1$; $K^{(\mathbf{0})} = 1$, $K^{(\mathbf{1})} = 0 - 1 \cdot 1/1 = -1$; $a_{\mathbf{2}} = 0 + 1 \cdot 2 = 2.000$; $P_{\infty,\mathbf{2}} = 1(1 - 1) = 0$; $P_{*,\mathbf{2}} = 1 \cdot 1 + 0 + 0.5 = 1.500$
- 2. $t = 2$: $v = 1.000$, $F = 2.500$, $K = 0.600$, $a_{\mathbf{3}} = 2.600$, $P_{\mathbf{3}} = 1.5(1 - 0.6) + 0.5 = 1.100$; $t = 3$: $v = -0.100$, $F = 2.100$, $K = 0.524$, $a_{\mathbf{4}} = 2.548$, $P_{\mathbf{4}} = 1.024$
- 3. Terms $-0.919$, $-1.577$, $-1.292$; $\ln L_d = -3.788$. The first term is $-\frac{\mathbf{1}}{\mathbf{2}} \ln 2\pi$ whatever the variances: $y_{\mathbf{1}}$ only fixes the level

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# A2: the concentrated likelihood [Proposed]

Solution discussed in class

## Task

- The data of A1, now with $\sigma_\eta^2 = 0.5\sigma_\varepsilon^2$ and $\sigma_\varepsilon^2$ unknown. Model: A1.
- 1. Run the filter with $\sigma_\varepsilon^2 = 1$ and write $F_t = \sigma_\varepsilon^2 F_t^*$.
- 2. Derive $\hat{\sigma}_\varepsilon^2$ by maximising the diffuse log-likelihood and compute it.
- 3. Compute the concentrated log-likelihood (without the diffuse term) and state how many parameters remain for the numerical search.
- Report: two ratios, one estimate, one number, one sentence.

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# A3: one backward-sampling step [Solved]

## Task

- Continue A1: after $y_2$ the filtered moments are $a_{2|2}$, $P_{2|2}$; suppose the draw for the next period is $\alpha_3 = 3.0$.
- 1. Compute $a_{2|2}$ and $P_{2|2}$ from A1.
- 2. Compute $G_2$, the conditional mean $m_2$ and variance $S_2$ of $\alpha_2|\alpha_3, y_1, y_2$.
- 3. Explain why $y_3$ is not needed in this conditional distribution.
- Report: two moments, three numbers, one sentence.

## Solution

- 1. $a_{2|2} = 2 + 0.6 \cdot 1 = 2.600$, $P_{2|2} = 1.5 - 1.5^2/2.5 = 0.600$
- 2. $G_2 = P_{2|2}/(P_{2|2} + \sigma_\eta^2) = 0.545$;
  $m_2 = 2.600 + 0.545(3.0 - 2.600) = 2.818$;
  $S_2 = P_{2|2}(1 - G_2) = 0.273$
- 3. Markov property: given $\alpha_3$, the future observations add nothing about $\alpha_2$; their information already entered through the draw of $\alpha_3$

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# A4: the inverse-gamma Gibbs block [Proposed]

Solution discussed in class

## Task

- A local level Gibbs sampler, $n = 50$; current state draw gives $\sum_t (y_t - \mu_t)^2 = 40$ and $\sum_t (\Delta\mu_t)^2 = 6.2$; priors $IG(2.5, 0.25)$ for both variances. Model: A3.
- 1. Write the two conditional posteriors with their parameters.
- 2. Compute their means and the implied $q$.
- 3. Explain why this sampler can never return $\sigma_\eta^2 = 0$, unlike ML.
- Report: two distributions, three numbers, one sentence.

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# A5: moments of the SV model [Solved]

## Task

- SV model with $\phi = 0.98$, $\sigma_\eta = 0.15$, Gaussian $\epsilon_t$.
- 1. Compute $\sigma_h^2$ and the kurtosis of $y_t$.
- 2. Compute $\mathrm{Corr}(y_t^2, y_{t-k}^2)$ for $k = 1, 10, 50$ and the half-life of a volatility shock.
- 3. Explain the difference from GARCH(1,1) with the same persistence.
- Report: two numbers, four numbers, one sentence.

## Solution

- 1. $\sigma_h^2 = 0.0225/0.0396 = 0.568$; kurtosis $3e^{0.568} = 5.30$
- 2. 0.173, 0.138, 0.054: slow decay from a low start; half-life $\ln 0.5 / \ln 0.98 = 34.3$ days
- 3. GARCH variance is known one step ahead; in SV it has its own shock, so excess kurtosis arises even with $\phi$ close to 1 and the likelihood needs an integral

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# A6: the QML measurement equation [Proposed]

Solution discussed in class

## Task

- The SV model of A5, written as $\ln y_t^2 = h_t + \xi_t$. Model: A5.
- 1. Give the mean and variance of $\xi_t$ and the signal-to-noise ratio $\sigma_h^2/\mathrm{Var}\xi_t$.
- 2. Compute $\mathrm{Corr}(\ln y_t^2, \ln y_{t-k}^2)$ for $k = 1, 10$ and name the ARMA model of $\ln y_t^2$.
- 3. Explain why the Kalman filter on $\ln y_t^2$ is consistent but inefficient.
- Report: three numbers, two numbers and a model, one sentence.

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# A7: one step of the bootstrap particle filter [Solved]

## Task

- SV model, $N = 4$ particles after propagation: $h^{(i)} = (-1; 0; 0.5; 1.5)$; observation $y_t = 2.0$; systematic resampling with $U = 0.3$.
- 1. Compute $w^{(i)} = N(y_t; 0, e^{h^{(i)}})$ and the normalised weights.
- 2. Compute the ESS and the likelihood increment $\hat{p}(y_t | Y_{t-1})$.
- 3. Resample: which particles survive, and how many times?
- Report: eight numbers, two numbers, one list.

## Solution

- 1. $w = (0.0029; 0.0540; 0.0924; 0.1206)$; $W = (0.011; 0.200; 0.342; 0.447)$
- 2. ESS $= 2.80$ of 4; $\hat{p} = \frac{1}{4} \sum w = 0.0675$, $\ln \hat{p} = -2.696$
- 3. $u = (0.075; 0.325; 0.575; 0.825)$ against $C = (0.011; 0.211; 0.553; 1)$: particles 2, 3, 4, 4; the low-volatility particle dies, the high-volatility one is copied twice

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# A8: auxiliary weights with an outlier [Proposed]

Solution discussed in class

## Task

- Local level, $\sigma_\varepsilon^2 = 0.5$, $\sigma_\eta^2 = 0.25$; equally weighted particles $x_{t-1}^{(i)} = (0; 1; 2; 3)$; outlier $y_t = 4.0$. Model: A7.
- 1. Compute the normalised first-stage weights $\propto N(y_t; x_{t-1}^{(i)}, \sigma_\varepsilon^2)$ and their ESS.
- 2. Compute the fully adapted weights $\propto N(y_t; x_{t-1}^{(i)}, \sigma_\varepsilon^2 + \sigma_\eta^2)$ and compare.
- 3. Explain what the bootstrap filter would do with the same outlier.
- Report: four weights and an ESS, four weights, one sentence.

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# A9: the Beveridge–Nelson cycle of an ARIMA(1,1,0) [Solved]

## Task

- Quarterly growth: $\Delta y_t - 0.8 = 0.35(\Delta y_{t-\mathbf{1}} - 0.8) + e_t$; this quarter $\Delta y_t = 2.0$.
- 1. Derive $\sum_{j \geq \mathbf{1}} \mathbb{E}_t(\Delta y_{t+j} - \mu)$ for an AR(1).
- 2. Compute the BN cycle $c_t$ and the long-run multiplier of $e_t$ on the trend.
- 3. Explain the sign of the cycle after a strong quarter.
- Report: one formula, two numbers, one sentence.

## Solution

- 1. $\mathbb{E}_t(\Delta y_{t+j} - \mu) = \phi^j(\Delta y_t - \mu)$, sum $= \frac{\phi}{\mathbf{1}-\phi}(\Delta y_t - \mu)$
- 2. $\frac{\phi}{\mathbf{1}-\phi} = 0.538$; $c_t = -0.538 \cdot 1.2 = -0.646$; multiplier $1/(1-\phi) = 1.538$
- 3. With positive growth persistence, a strong quarter raises expected future growth: the trend jumps by more than output, so output is *below* trend

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# A10: local level, IMA(1,1) and Beveridge–Nelson [Proposed]

Solution discussed in class

## Task

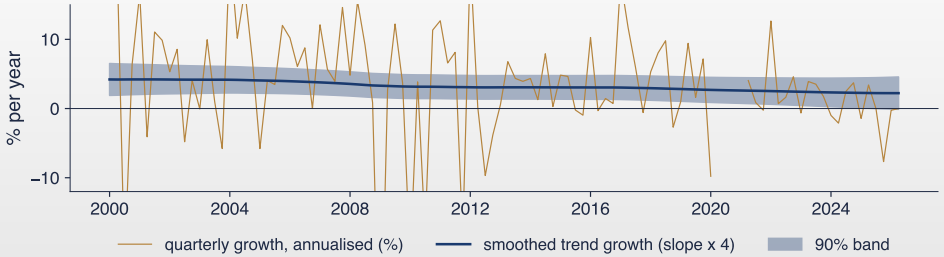
- Local level with $\sigma_\varepsilon^2 = 1$, $q = 0.5$. Model: A9.
- 1. Find the MA coefficient $\theta$ of $\Delta y_t$.
- 2. Solve the steady-state Riccati equation $P = P - P^2/(P+1) + q$ and give the gain $k = P/(P+1)$.
- 3. Show that the BN trend of the IMA(1,1), $y_t + \theta u_t$, equals the steady-state filtered level $a_{t|t}$.
- Report: one number, two numbers, one derivation.

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# B1: Romanian trend growth [Solved]

- **Question**: has Romania's trend growth fallen since the 2000s, once the uncertainty of the estimate is taken into account?
- **Inputs**: $100\times$log real GDP (Eurostat, SCA), 2000Q1–2026Q2, $T = 106$; 2020Q2–Q4 treated as missing
- **Tasks**
  1. Estimate the local linear trend by exact diffuse ML with the numpy filter and with statsmodels; compare the parameters and the log-likelihoods.
  2. Smooth the slope and report the annualised trend growth $4\hat{\nu}_t$ with a 90% band in 2005, 2012, 2019 and in the last quarter.
  3. Check whether any variance is at the boundary.
  4. Interpretation: is the fall in trend growth statistically meaningful?
- **Report**: two parameter sets, four growth rates with bands, one sentence
- Notebook: section B1

# B1: solution [Solved]




- numpy: $\sigma_\varepsilon^2 = 0.6013$, $\sigma_\xi^2 = 3.9577$, $\sigma_\zeta^2 = 0.0039$, $\ln L_d = -231.08$; statsmodels: 0.6011, 3.9581, 0.0039, −229.24 (the difference in $\ln L$ is $\ln 2\pi$, two diffuse steps)
- Trend growth: 4.1% (s.d. 1.1) in 2005, 3.1% (1.0) in 2012, 2.8% (1.1) in 2019, 2.2% (1.4) at the end
- Interpretation: a fall is suggested but not established
  - the slope variance is small, so trend growth moves slowly
  - the bands overlap widely; level shocks absorb the 2009 recession

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# B2: TVP regressions for Hungary and Poland [Proposed]

- **Question**: has the co-movement of Hungarian and Polish inflation with euro-area inflation changed over time?
- **Inputs**: quarterly HICP inflation (% a.r.), 2005Q4 onward; euro-area inflation net of its seasonal pattern; the TVP regression of the primer slide Local trends and time-varying coefficients; model: B1
- **Tasks**
  1. Estimate the TVP regression by exact diffuse ML (five diffuse states) for each country.
  2. Report $\hat{\sigma}_b^2$ and the smoothed $b_t$ with standard errors in 2008Q3, 2019Q4 and 2022Q4.
  3. Compute the LR test of $\sigma_b^2 = 0$ and its $\frac{1}{2}\chi_1^2$ $p$-value; say why this $p$-value is only indicative.
  4. Interpretation: is the slope higher than one, and is the co-movement stable or episodic?
- **Report**: a table of two rows, six slopes with standard errors, two tests, one sentence
- Notebook: section B2

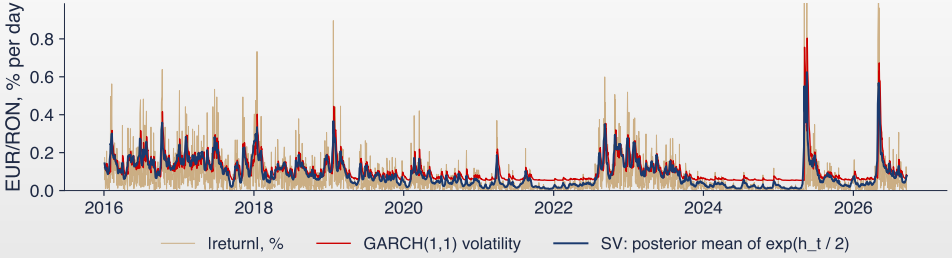
Back Exercise list

# B3: stochastic volatility of EUR/RON [Solved]

- **Question**: how persistent is the volatility of the leu against the euro, and how does SV compare with GARCH?
- **Inputs**: daily log returns of the BNR EUR/RON reference rate, %, demeaned, 2016–2026, $T = 2686$; KSC Gibbs sampler with $c = 10^{-3}\mathrm{Var}(y)$
- **Tasks**
  1. Run the KSC Gibbs sampler and report posterior means and 90% intervals of $\mu, \phi, \sigma_\eta$ with inefficiency factors.
  2. Estimate a Gaussian GARCH(1,1) on the same data.
  3. Plot the posterior mean of $\exp(h_t/2)$ against the GARCH volatility.
  4. Interpretation: what does the half-life of volatility say about a managed exchange rate?
- **Report**: a table of three parameters, three GARCH numbers, one chart, one sentence
- Notebook: section B3

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# B3: solution [Solved]




- $\phi = 0.957$ [0.944; 0.969], $\sigma_\eta = 0.541$ [0.488; 0.597], $\mu = -5.628$ [$-6.032$; $-5.210$]; inefficiency factors 18, 42, 1 (20 000 draws)
- GARCH(1,1): $\alpha = 0.225$, $\beta = 0.753$; daily s.d. 0.121%, kurtosis 27.8
- Interpretation: the half-life of a log-volatility shock is about 16 days
  - quiet periods with rare bursts are what a managed float produces
  - the high kurtosis says that Gaussian $\epsilon_t$ is not enough (B4)

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## B4: the particle filter for the EUR/RON SV model [Proposed]

- **Question**: is SV better than GARCH for EUR/RON on the likelihood scale, and is the particle likelihood reliable here?
- **Inputs**: the data and the posterior means of B3; bootstrap particle filter with systematic resampling; model: B3 and A7
- **Tasks**
  1. Run the filter with $N = 500, 2000, 10\,000$ particles, 30 times each, and report the s.d. of $\ln \hat{L}$.
  2. Report the minimum and median ESS and the date of the minimum.
  3. Compare $\ln \hat{L}$ (large $N$) with the GARCH maximum.
  4. Interpretation: what does the ESS collapse tell you about the model?
- **Report**: three standard deviations, two ESS values and a date, two log-likelihoods, one sentence
- Notebook: section B4

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# B5: the Romanian output gap in real time [Proposed]

- **Question**: how much does the end-of-sample estimate of the Romanian output gap change when later data arrive?
- **Inputs**: the data of B1; smooth-trend UC with AR(2) cycle (`fit_smooth_trend`); quasi-real time: re-estimate on data up to each quarter from 2012Q4 (parameters every 4 quarters) and keep the filtered gap; model: B1 and A10
- **Tasks**
  1. Compute the series of real-time (filtered, end-of-sample) gaps and the final smoothed gap.
  2. Compute the Orphanides–van Norden statistics: correlation, noise-to-signal ratio (s.d. of revisions over s.d. of the final gap), share of opposite signs.
  3. Report the largest revision.
  4. Interpretation: can monetary policy rely on the current estimate of the gap?
- **Report**: four statistics, one number, one sentence
- Notebook: section B5

# C1: is Romanian trend inflation anchored? [Proposed]

- **Question**: is the trend of Romanian inflation inside the BNR target band (2.5% $\pm$ 1 pp), and how robust is the answer?
- **Inputs**: quarterly HICP inflation (% a.r.); UC-SV of Stock and Watson (2007) with quarterly seasonal dummies (`ucsv_gibbs`); model: B3
- **Tasks**
  1. Write a five-line pre-registration: hypothesis, data, the anchoring statistic, the robustness grid, what would falsify the claim.
  2. Estimate the model for start dates 2001 and 2006 and for $\gamma = 0.1$ and 0.2; report the trend in the last quarter with a 68% band.
  3. Compute the posterior probability that the trend lies in 1.5–3.5% in the last quarter and in 2019Q4.
  4. Interpretation: what can and cannot be concluded, and how would you turn this into a project?
- **Report**: a five-line plan, a table of four rows, a project plan
- Notebook: section C1

# C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise state space estimation. The answer:
- `(a) The Kalman filter is useless if the errors are not Gaussian.`
- `(b) For a random-walk state, set P1 = 10^10 and keep all n terms of the likelihood.`
- `(c) An ML estimate of the level variance equal to zero proves that the level is constant.`
- `(d) SV and GARCH are the same model, since both make volatility persistent.`
- `(e) The particle-filter likelihood is unbiased, so its logarithm is unbiased too.`
- `(f) The Durbin-Koopman simulation smoother needs only one state-smoother pass per draw.`
- `(g) With an AR(1) cycle, the correlation between trend and cycle shocks can be estimated.`
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, a number from today's tasks (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

## Key takeaways

- The exact diffuse filter lets the first observations fix the nonstationary states; they carry no information about the variances
- Sampling the states is the Kalman machinery run backwards (FFBS) or once more on simulated data (DK)
- SV adds a shock to the variance; particle filters estimate its likelihood, unbiased in levels only
- Trend growth, gaps and anchored trends are estimates with wide bands: report them, and report real-time values
- An AI answer is a draft: check the initialisation, the boundary, the log-likelihood bias and the identification

## After the seminar

- Lecture 6 develops each topic of today: the exact diffuse filter, the likelihood at the boundary, the three samplers, KSC, particle filters and PMMH, TVP, DFM, MNZ and UC-SV
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: trend inflation and anchoring in Central and Eastern Europe
- Reading: Durbin and Koopman (2012), Ch. 4–5; Kim, Shephard and Chib (1998); Morley, Nelson and Zivot (2003); Stock and Watson (2007)
- **The seminar is for practice and is not graded**

# References (1/2)


- Beveridge, S., & Nelson, C. R. (1981). A new approach to decomposition of economic time series into permanent and transitory components with particular attention to measurement of the 'business cycle'. *Journal of Monetary Economics*, 7(2), 151–174. doi:10.1016/0304-3932(81)90040-4
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. doi:10.1016/0304-4076(86)90063-1
- Carter, C. K., & Kohn, R. (1994). On Gibbs sampling for state space models. *Biometrika*, 81(3), 541–553. doi:10.1093/biomet/81.3.541
- Durbin, J., & Koopman, S. J. (2002). A simple and efficient simulation smoother for state space time series analysis. *Biometrika*, 89(3), 603–616. doi:10.1093/biomet/89.3.603
- Durbin, J., & Koopman, S. J. (2012). *Time Series Analysis by State Space Methods* (2nd ed.). Oxford University Press. doi:10.1093/acprof:oso/9780199641178.001.0001
- Gordon, N. J., Salmond, D. J., & Smith, A. F. M. (1993). Novel approach to nonlinear/non-Gaussian Bayesian state estimation. *IEE Proceedings F (Radar and Signal Processing)*, 140(2), 107–113. doi:10.1049/ip-f-2.1993.0015
- Harvey, A., Ruiz, E., & Shephard, N. (1994). Multivariate stochastic variance models. *The Review of Economic Studies*, 61(2), 247–264. doi:10.2307/2297980
- Kim, S., Shephard, N., & Chib, S. (1998). Stochastic volatility: Likelihood inference and comparison with ARCH models. *The Review of Economic Studies*, 65(3), 361–393. doi:10.1111/1467-937X.00050

# References (2/2)


- Koopman, S. J. (1997). Exact initial Kalman filtering and smoothing for nonstationary time series models. *Journal of the American Statistical Association*, 92(440), 1630–1638. doi:10.1080/01621459.1997.10473685
- Morley, J. C., Nelson, C. R., & Zivot, E. (2003). Why are the Beveridge–Nelson and unobserved-components decompositions of GDP so different? *The Review of Economics and Statistics*, 85(2), 235–243. doi:10.1162/003465303765299765
- Orphanides, A., & van Norden, S. (2002). The unreliability of output-gap estimates in real time. *The Review of Economics and Statistics*, 84(4), 569–583. doi:10.1162/003465302760556422
- Pitt, M. K., & Shephard, N. (1999). Filtering via simulation: Auxiliary particle filters. *Journal of the American Statistical Association*, 94(446), 590–599. doi:10.1080/01621459.1999.10474153
- Shephard, N., & Harvey, A. C. (1990). On the probability of estimating a deterministic component in the local level model. *Journal of Time Series Analysis*, 11(4), 339–347. doi:10.1111/j.1467-9892.1990.tb00062.x
- Stock, J. H., & Watson, M. W. (1998). Median unbiased estimation of coefficient variance in a time-varying parameter model. *Journal of the American Statistical Association*, 93(441), 349–358. doi:10.1080/01621459.1998.10474116
- Stock, J. H., & Watson, M. W. (2007). Why has U.S. inflation become harder to forecast? *Journal of Money, Credit and Banking*, 39(s1), 3–33. doi:10.1111/j.1538-4616.2007.00014.x